



International Workshop on Photonics and Optical Materials (PHOTOMAT - 2026)



Organized by

Dept. of Applied Physics,
Defence Institute of
Advanced Technology,
Girinagar, Pune - 411025,
MH, India



October 7th - 9th, 2026



Takshshila Auditorium

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WORKSHOP SCHEDULE

International Workshop on Photonics and Optical Materials (PHOTOMAT-2026)

All times are in IST (+5:30 GMT)

Day 1: PHOTONICS (07 October 2026)		
Time	Training Contents	Lecturer/Instructor
9:00 am – 9:30 am	Registration and Inauguration	
9:30 am – 10:30 am	Plenary Lecture	Dr. Jagannath Nayak, Director General, MSS, DRDO
10.30 am – 10:45 am	High Tea Break	
10:45 am – 11:45 am	Design and Development of Ultrafast Yb-doped Fiber Oscillators and Amplifiers in Diverse Operating Regimes	Dr. P. K. Mukhopadhyay, SO-H, ALDAD, RRCAT.
11:45 am – 12:15 pm	All-Fiber Ultrafast Laser System for DEW Applications	Dr. Shyamal Mondal, Asst. Prof., Dept. of AP, DIAT, Pune
12.15 pm – 12.45 pm	Physical Demonstration by Industry	Revine Tech (Accelerating Innovation)
1:00 pm – 1:30 pm	Laboratory Session	Dr. A.V.R. Murthy, Dr. Shyamal Mondal, Mr. Pramod Bankar, Mr. Vikrant Gurmitkal
1:30 pm – 2:30 pm	Lunch Break	
2:30 pm – 5:00 pm	Laboratory Session	Dr. A.V.R. Murthy, Dr. Shyamal Mondal, Mr. Pramod Bankar, Mr. Vikrant Gurmitkal
5:00 pm – 6:00 pm	Optical Communication Technologies from Classical to Quantum: Challenges and Roadmap	Prof. Shailendra K. Varshney, Electronics and Electrical Communication Engg., IIT Kharagpur
6:00 pm – 7:00 pm	Building a long-distance quantum repeater network using atomic quantum Interconnects	Prof. Eden Valentin Figueroa Barragan, Director of Stony Brook, Quantum Institute, NY, USA
7:30 pm – 9:00 pm	Gala Dinner	
Day 2: OPTICAL WIRELESS COMMUNICATION AND UNDERWATER TESTING (08 October 2026)		
9:00 am – 10:00 am	Emerging Paradigms in Inertial Sensing: Overcoming Limits via Quantum Photonics	Sree Ramana Mukkamala Scientist H, DRDO-RCI, Hyderabad



10:00 am – 10:45 am	Cooperative Source Localization of Underwater Plumes using a Team of UUVs	Dr. Rakesh R. Warriar, Asst. Prof., NIT Calicut
10:45 am – 11:00 am	Tea Break	
11:00 am – 12:00 pm	Photonic Techniques for Generation, Distribution, and Measurement of Microwave Signals	Prof. Deepa Venkitesh, Dept. of EE, IIT Madras
12:00 pm – 12:30 pm	Underwater Vision Based Surveillance and Tracking using Machine Learning	Dr. Pooja Agarwal, Asst. Prof. Dept. of ME, DIAT, Pune
12:30 pm – 1:00 pm	Laser-Induced Photoacoustic Underwater Communication at 1 kbps Using a 1080 nm High- Power Fiber Laser	Prof. Devnath Dhirhe, Dept. of AP, DIAT, Pune
1:00 pm – 1:30 pm	Underwater Optical Communication System Design from Basic Building Blocks to Prototype	Dr. A.V.R. Murthy, Asst. Prof., Dept. of AP, DIAT, Pune
1:30 pm – 2:30 pm	Lunch Break	
2:30 pm – 6:30 pm	Laboratory Session	Dr. A.V.R. Murthy, Dr. Shyamal Mondal, Mr. Pramod Bankar, Mr. Vikrant Gurmitkal
Day 3: OPTICAL MATERIALS (09 October 2026)		
9:00 am – 10:00 am	The Organic-Inorganic Hybrid and Halide Perovskites: The Highly Tuneable New Breed of Optical Materials	Prof. Satishchandra B. Ogale, Director, RISE, TCG-CREST, Kolkata
10:00 am – 10:45 am	Terahertz Metasurface: A viable platform for exploring bound states in the continuum	Prof. Dibakar Roy Chowdhury, Anurag University, Hyderabad
10:45 am – 11:00 am	High Tea Break	
11:00 am – 11:45 am	Harnessing optical nonlinearities for ultrashort laser pulse applications	Dr. Aditya Dharmadhikari, Sc. Officer F, DNAP, TIFR, Mumbai
11:45 am – 12:15 pm	Carrier Injection as a versatile mechanism for optical switching and beam steering	Dr. Sooraj Ravindran, Asso. Prof., IIST, Trivandrum



12:15 pm – 1:00 pm	Innovative Design and Fabrication of Specialty Fibers for Enhanced Fiber Laser Performance	Dr. Anirban Dhar, Principal Scientist, FOPD, CGCRI, Kolkata
1:00 pm – 1:30 pm	Halide Perovskite Heterostructures for Photodetector Applications	Prof. Tejashree Bhawe, HoD, Dept. of AP, DIAT, Pune
1:30 pm – 2:30 pm	Lunch Break	
2:30 pm – 3:00 pm	Approaches at the interface of materials and optics: sensors and device development	Sr. Prof. Sangeeta Kale, Dept. of AP, DIAT, Pune
3:00 pm – 3:30 pm	Quartz-enhanced photo-acoustic spectroscopy (QEPAS) for detection of H ₂ S and NH ₃	Prof. Suwarna Datar, Dept. of AP, DIAT, Pune
3:30 pm – 4:00 pm	Valedictory Session	
4:00 pm – 4:30 pm	Evening Tea & Snacks	

Chief Guest

Dr. Jagannath Nayak
Director General, MSS, DRDO

Guest of Honor

Dr. Gaurav Singhal
Director, CHESS, DRDO Hyderabad

Chair Person

Prof. Tejashree Bhawe
HoD & Professor, DIAT

Convenor

Dr. Shyamal Mondal
Asst. Professor, DIAT

Co-Convenor

Dr. A V R Murthy
Asst. Professor, DIAT

Organizing Committee

Prof. Tejashree Bhawe	HoD & Professor, DIAT
Prof. Sangeeta Kale	Sr. Professor, DIAT
Prof. Suwarna Datar	Professor, DIAT
Prof. Devnath Dhirhe	Professor, DIAT
Dr. Shyamal Mondal	Asst. Professor, DIAT
Dr. A V R Murthy	Asst. Professor, DIAT
Dr. Pooja Agarwal	Asst. Professor, DIAT